
Table as Prompt: An Interactive Guide to Tabular Foundation Models

Jiarui Xu*

Tianming Zhou

Tommy Tai

Joshua Pan

Alexander Statnikov

Nitesh Kumar

Affirm, Inc.

Abstract

Tabular prediction has long relied on fitting and tuning a separate model for each dataset. Prior-data fitted networks (PFNs) instead pretrain on sampled tasks, then use labeled rows as context to predict queries with fixed weights. Our browser guide teaches this transition through TabICLv2 and nanoTabICL. An interactive structural causal model (SCM) illustrates synthetic task generation; diagrams connect the PFN objective, TabICLv2 training, and frozen inference; and synchronized visualizations compare linear, tree, k -nearest-neighbor, and in-context predictions. A real exported checkpoint runs locally in the browser, while a self-guided notebook provides a measured case study. The website needs no installation or inference service; the notebook installs TabICL locally.

1 The concept: tabular in-context learning

Many conventional tabular workflows train, and often tune, a separate model for each dataset. PFN-style tabular foundation models instead pretrain a shared predictor across sampled tasks. At inference, labeled rows form the context and unlabeled rows are queries; the model predicts without task-specific gradient updates (Müller et al., 2022). TabPFN provided an early demonstration of this approach for small tabular classification tasks (Hollmann et al., 2023).

This adaptation is **tabular in-context learning**. For PFN-style models, the sampled-task prior and expected query-loss objective shape the learned inductive bias; architectures and inference procedures differ across models. TabICLv2 follows TabICL’s feature processing, row compression, and dataset-wise ICL design, while adding repeated feature grouping and scalable attention (Qu et al., 2026). Its feature groups depend on relative column positions, and its evaluated inference procedure ensembles random column shuffles and preprocessors.

The TabICLv2 paper also provides nanoTabICL, a compact educational implementation of this architecture. We use it to trace tensor shapes and data movement through the forward pass. These are model-specific design choices, not properties shared by all tabular foundation models; our diagrams illustrate computation but do not pretrain a model in the browser.

Tabular foundation models remain at an early stage. Recent systems span small and large tables and both classification and regression, while leaving open questions about robustness, scale, efficiency, data regimes, and evaluation. This trajectory suggests the potential for a reusable intelligence layer for structured data rather than another pipeline rebuilt for every dataset. Our tutorial aims to help researchers and industry practitioners understand this transition, evaluate its limits against strong task-specific baselines, and build on it.

*Correspondence to: Jiarui Xu, jiarui.xu@affirm.com.

2 Audience and prerequisites

Intended audience. The tutorial is designed for advanced undergraduate and graduate students, researchers, and industry practitioners. A basic understanding of supervised machine learning is sufficient; no prior familiarity with tabular foundation models is assumed. The interactive website runs in a modern browser without specialized hardware or a remote inference service.

3 Learning objectives

After completing the materials, a learner will be able to:

- **Explain** task-specific fitting, adaptation through context, and the PFN objective.
- **Trace** tensor shapes through nanoTabICL and relate them to TabICLv2.
- **Compare** linear, tree, nearest-neighbor, and foundation-model predictions.

4 Linked papers (2022–2026)

These papers establish the concept and its current frontier:

1. Müller et al. *Transformers Can Do Bayesian Inference*. ICLR 2022. Introduces PFNs.
2. Hollmann et al. *TabPFN: A Transformer That Solves Small Tabular Classification Problems in a Second*. ICLR 2023.
3. Hollmann et al. *Accurate Predictions on Small Data with a Tabular Foundation Model*. Nature, 2025.
4. Qu et al. *TabICL: A Tabular Foundation Model for In-Context Learning on Large Data*. ICML 2025.
5. Qu et al. *TabICLv2: A Better, Faster, Scalable, and Open Tabular Foundation Model*. ICML 2026.

5 Teaching materials summary

The archive contains the guide, a self-guided notebook, and source code:

Website: <https://affirm.github.io/tabular-foundation-models-tutorial/>

- **Problem and adaptation.** UCI Online Shoppers demonstrates mixed feature types and class imbalance; masking and subsampling illustrate missing and limited context. Learners contrast task-specific fitting with adaptation through labeled examples.
- **PFN theory and training.** An interactive SCM generates illustrative tasks, not the full TabICLv2 prior. Context/query splits and query loss connect synthetic tasks to the PFN objective, while diagrams explain the TabICLv2 recipe without claiming to train in the browser.
- **Frozen inference.** A real browser checkpoint produces eight-view TabICLv2 predictions. An inspector traces one selected view through nanoTabICL tensor shapes and attention; compact contexts are labeled as out-of-regime illustrations.
- **Browser comparison.** Four synchronized canvases compare logistic regression, a tree, k -nearest neighbors, and TabICLv2 on the same context and query. This is an illustration, not a tuned benchmark.
- **Notebook and references.** A real-table case study compares prespecified XGBoost with frozen TabICLv2 weights using 300 context rows. A reference collection pairs models and benchmarks with their papers, code, data, weights, and documentation.

We hope this tutorial provides researchers and industry practitioners with an accessible foundation for understanding and advancing the field. We welcome feedback, discussion, and collaboration on future educational resources and research.